



Birla Institute of Technology & Science, Pilani

Pilani Campus

AUGSD/AGSRD Division

FIRST SEMESTER 2025-2026

Course Handout

Date 01/08/2026

Course No: MEL G625

Course Title: Advance Analog and Mixed Signal Design

Instructor-in- Charge: Neeraj Mishra

- 1. Course description:** Mixed signal blocks design issues,, design of high resolution, speed Comparators,, design of Active Mixers-balanced/ unbalanced,, Data Converters Architectures, CMOS multipliers , dividers and modulators , filters-active/ passive/ switched capacitor, Signal Conditioning and sensor associated circuits, Frequency Synthesizers, Phase Locked Loop, Power Management Circuits, Integrated Voltage Regulators, Energy Harvesting Circuits, Analog Testing, Analog layouts, noise issues, emerging trendshigh
- 2. Scope and Objective:** This course is designed to provide in-depth knowledge and hands-on expertise in designing advanced analog and mixed-signal integrated circuits with a focus on modern CMOS technologies. The course aims to bridge the gap between analog fundamentals and practical design of performance-critical blocks such as Phase-Locked Loops (PLLs), oscillators, and Analog-to-Digital Converters (ADCs). It equips students to analyze, architect, and implement key mixed-signal blocks considering design trade-offs, non-idealities, and layout constraints. Emerging trends such as energy harvesting, sensor interface circuits, and digitally assisted analog systems are also discussed to build readiness for research and industrial innovation.

3. Course learning outcomes:

Upon successful completion of this course, the students will be able to:

- a. Analyse and Design** high-resolution, high-speed comparators, mixers, and signal conditioning blocks for mixed-signal interfaces.
- b. Architect and Evaluate** various ADC and DAC topologies based on speed, resolution, power, and area metrics.
- c. Design and Simulate** PLLs and oscillator-based clock generation systems with attention to phase noise, jitter, and loop stability.
- d. Implement of Biasing Circuits and Optical/Wireline Receiver Transmitter circuits.**



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- e. **Understand and Apply** analog layout methodologies, matching principles, and noise-aware design techniques.
- f. **Demonstrate Proficiency** in using industry-grade tools (e.g., Cadence) to model, simulate, and layout mixed-signal circuits.

4. Text books and references:

There is no single reference for this. It will be based on multiple journal articles and books that will be shared along with the relevant lectures.

TB1. RF Microelectronics, B. Razavi, 2nd Ed

BRB1. Design of CMOS phase-locked loops: from circuit level to architecture level, Behzad Razavi, 2020/1/30, Cambridge University Press

RB1. CMOS Mixed Signal Circuit Design, R. Jacob, Baker, IEEE Press.

RB2. Analog-to-Digital Conversion, 4th Edition *Marcel J.M Pelgrom*.

RB3: CMOS Circuit Design, Layout and Simulation – R. J. Baker, 3rd Edition, Wiley-IEEE.

RB4: Analog Design Essentials Hardcover – 31 May 2006 by Willy M. C. Sansen.

RB5: IEEE ISSCC/CICC/AISSCC/ESIRCC conferences papers and IEEE JSSC/TCAS-I/II, research papers related to topics.

5. Method of conduct of the course

This course follows a **design-centric and research-driven pedagogy**, aimed at cultivating both analytical depth and practical proficiency in analog and mixed-signal design.

- **Lecture Sessions** will focus on conceptual clarity, design trade-offs, and architecture-level intuition using real-world mixed-signal systems as case studies. Select sessions will be used for dissecting key blocks such as PLLs, ADCs, comparators, mixers, and voltage regulators.
- **Research Paper Discussions:** Students will regularly read, present, and critique seminal and recent research papers from IEEE JSSC, TCAS, and ISSCC to stay aligned with state-of-the-art trends. Discussions will emphasize circuit innovation, design methodologies, and practical constraints.
- **Hands-on Lab Sessions** will be conducted using industry-standard EDA tools (e.g., Cadence, Spectre, Virtuoso) to implement and simulate analog and mixed-signal building blocks. Students will perform schematic-level design, corner simulations, layout practices, and post-layout verifications.



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- **Mini Projects and Design Challenges:** Learners will undertake short design assignments focused on specific circuit blocks (e.g., comparator, VCO, LDOs BGRs), progressively leading to a final design project.
- **Collaborative Learning** will be encouraged through team-based design reviews, lab debugging, and peer feedback on presentations and reports.

This method ensures that students develop a **deep understanding of design principles** while also acquiring **practical experience in implementing and validating real circuits**, preparing them for research, industry, or further academic pursuits.

6. Course Plan:

Lecture/ Topics

S.No.	Module	Lecture	Learning Outcomes Students are expected to know following concepts:
1	Mixed-Signal Design Foundations	(1 Lecture)	• Introduction to Mixed Signal Design • Overview of Analog vs Digital Blocks • Key Performance Metrics • Design Flow and Specifications
2	Basics of Amplifier stability and Mismatch	3 Lectures	• Bode Plot and Design of OTAs • Mismatch in Analog circuits
3	Biasing Circuits	4 Lectures	• Current Mirrors, BMR Circuits, LDO, BGR
4	Fundamentals of Phase Noise and Jitter, Oscillators Fundamentals	6 Lectures	• Noise, power spectral density, thermal and flicker noise, jitter definitions, jitter characterization, relationship between phase noise & jitter, • Phase noise analysis of VCO, inverter, delay lines; jitter in delay lines vs. ring oscillators. Oscillator Fundamentals • Ring oscillators, VCOs: Tuning linearity, tuning schemes, Delay cells: replica biasing, Power supply sensitivity, improved linearity and symmetric delay loads, Large swing VCOs, Supply-regulated VCOs • Differential ring VCOs with large swing, pseudo-differential delay cells, VCO Buffers



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5	Phase-Locked Loops	5 Lectures	- Type-I PLLs: VCO Phase Alignment; Dynamics of Type-I PLL; Frequency Multiplication; Drawbacks of Type-I PLLs - Type-II PLLs: Phase/Frequency Detectors; Charge Pumps; CP PLLs; Transient Response - PLL Non-Idealities: PFD/CP Non idealities; Circuit Techniques; VCO Phase Noise; Reference Phase Noise
6	Basics of Wireline & Optical Transceivers and Analog Layout	3 Lectures	- Design of Optical Receivers and Wireline Transmitter Circuits - Layout techniques of analog circuits
7	Comparators and sampling Circuits	6 Lectures	- High-Speed , high-resolution comparators - Bootstrapping circuit, Comparator Design in ADCs
8	ADC Architectures	7 Lectures	- ADC Overview & Performance Metrics - Pipelined ADCs, - SAR ADCs - Oversampling DCs
9	DAC Architectures	7 Lectures	- Introduction to to DAC - R2R , C2C DAC - Current Steering DAC
9	Basics/Overview of Mixers, Modulators and interface circuits	1/2 Lectures	- Active and Passive Mixers - I/Q Modulators and Demodulators - CMOS Modulators
	Total	44	

7. Evaluation Scheme:

Eval. Component	Duration	Weightage	Date & Time	Nature
Mid-Term Exam	90 minutes	25%		OB/ CB
Reading assignments/ Design Project/ Final Project/ Quiz/ Presentations/ Paper submission	Continuous	40%	Spread throughout the semester	OB
Comprehensive Exam	3 Hrs	35 %		CB/OB



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8. Design Project & Assignments: A series of reading assignments and/or design-based mini-projects will be given throughout the semester. Students are expected to submit periodic progress reports. The culmination of the project will include a final project report (based on implementation using Cadence or equivalent EDA tools), an oral presentation (viva), and a technical paper submission in IEEE format.

9. Instructor Consultation Hours: Consultation hours will be announced in class. Students may also schedule a one-on-one meeting by contacting the instructor via email.

10. Notices & Announcements: All important course-related announcements will be posted on the EEE Department Notice Board and uploaded to the NALANDA LMS portal.

11. Make-up Policy: Make-up assessments will be considered only for valid and documented reasons. Prior approval from the Instructor-in-Charge (IC) is mandatory.

12. Important Note on Lab Participation: Active participation in lab sessions is essential. A significant portion of the course evaluation is based on the student's ability to design and implement circuits accurately and efficiently. Exam questions will primarily focus on design principles and conceptual understanding.

Instructor-in-Charge

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